

Split Complex Lorentzian Dynamics

Part XXVI: Varying Metric-Oriented Fibres, Algebraic Frame Trivializations, and the
Topological Frontier

From the closed one-fibre torsor to transition data over an arbitrary base, with the topology of
variation isolated for Part XXVII

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Abstract

Part XXV closed the algebraic one-fibre problem. A positively oriented orthonormal frame carrier had been reconstructed intrinsically, the visible group acted freely and transitively on it, the lifted carrier carried a free and transitive action of the internal group, and the two levels were connected by an equivariant two-sheeted projection with kernel $\{\pm 1\}$. The remaining gap to a spin structure could therefore no longer be blamed on an unidentified internal group, an unresolved sign, or an incomplete frame carrier. The missing object had to be global: a coherent family of such fibres over a base.

Task 26 begins exactly there and deliberately does not assume a vector bundle. For an arbitrary base type B and a dependent family $E : B \rightarrow \text{Type}$, it introduces the minimal metric-oriented rank-three input used in the formal run: each E_b is a real inner-product space of finite rank three, together with an orientation o_b . From these data it reconstructs the positively oriented orthonormal frame carrier $\mathcal{F}_b^+$ in every fibre, the total set $\mathcal{F}_{\text{tot}}^+ = \sum_{b:B} \mathcal{F}_b^+$ and its projection to B , and the intrinsic orientation-preserving isometry group G_b^+ of each fibre. The action $G_b^+ \curvearrowright \mathcal{F}_b^+$ is proved free and transitive, with a unique transporter between any two frames.

A fixed visible group does not arise canonically from the fibre family. After choosing an orientation-preserving metric identification $\phi_b : E_b \simeq E_0$ with a model space, one obtains an isomorphism $G_b^+ \simeq G_0^+$. Changing ϕ_b changes this group identification by conjugation with the corresponding transition element. Thus a constant model group is available only through presentation data, not as a definitional identification of all fibres.

The strongest closed notion of local triviality in Task 26 is intentionally algebraic. Over any subset $U \subseteq B$, a Level-1 trivialization is a family of orientation-preserving linear isometries $\phi_b : E_b \simeq E_0$. Such trivializations exist over every U by classical choice, hence in particular over the whole base. This is a decisive boundary: set-theoretically the family is globally trivializable, so no geometric obstruction can live at Level 1. Each fibre trivialization induces a full frame-space equivalence over U , while the exact reverse reconstruction proved by the task is the sharper statement that orientation-preserving identifications with the model are equivalent to choices of one positively oriented orthonormal frame. For two trivializations, the overlap transformation $g_{ij}(b) = \phi_{j,b} \phi_{i,b}^{-1}$ lies in G_0^+ and satisfies identity, inverse, and triple-overlap laws.

The new difficulty appears precisely when one asks for topology. The automatically available sigma topology on $\sum_b E_b$ is formally shown to be unsuitable: every fibre is open, and every algebraic fibrewise trivialization is automatically continuous for it. It therefore encodes no coherence of the variation in b . Task 26 does not invent a topology to force the desired conclusion. No topological vector bundle, frame bundle, principal bundle, lifted family, or spin structure is constructed. The formal run consists of nine new modules and 1615 lines, exposes 46 audited endpoints, reports a focused green Task-26 build and a whole-project green build of 8298 jobs with zero errors, and contains no proof hole or prohibited proof escape [2].

The established comparison is now exact. Classical bundle theory distinguishes a mere family of isomorphic vector spaces from a locally trivial topological vector bundle; an oriented metric rank-three bundle then has an oriented orthonormal frame bundle whose transition maps take values in the orientation-preserving isometry group [7, 3, 4]. Task 26 has reconstructed the entire pointwise and algebraic precursor of that picture, including transition laws, but not the topology that makes the transition functions continuous and the total frame carrier locally a product. Likewise, the already reconstructed one-fibre double structure is still not a spin structure: the latter is a global equivariant lift of the oriented orthonormal frame bundle, not merely a fibrewise two-level torsor [5].

Part XXVII should therefore not begin by lifting the algebraic transition maps. It should first determine the correct topology of variation: whether it must be supplied as additional total-space data or can be reconstructed from a compatible atlas of local product charts; how continuity of fibre operations, frame charts, and transition transformations is expressed; and which equivalence between topological local triviality and atlas data can actually be proved. Only once this Level-2 structure exists will it be meaningful for a later stage to ask whether the already certified two-sheeted internal projection can be made compatible with the transition system over the base. The Conservation of Difficulty has moved from internal algebra to global topological coherence.

Keywords: varying metric-oriented fibres; oriented orthonormal frames; local trivialization; transition transformations; frame family; structural group; topology; frame bundle; spin structure; Lean 4; formal verification; Conservation of Difficulty.

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1 After the one-fibre closure: what is still missing?

The previous stage ended with a complete algebraic diagram on a single metric-oriented three-dimensional carrier [6, 1]. Denoting its ordinary positively oriented orthonormal frames by $\mathcal{F}^+(E)$, the visible group by G_{vis} , the lifted frame carrier by $\tilde{\mathcal{F}}$ and the internal group by $\mathcal{L}$, the formal results may be compressed to

$$\begin{array}{ccc} \mathcal{L} & \curvearrowright & \tilde{\mathcal{F}} \\ \downarrow p & & \downarrow \pi \\ G_{\text{vis}} & \curvearrowright & \mathcal{F}^+(E), \end{array} \quad (1)$$

where both horizontal actions are free and transitive and π is equivariant along p . The kernel of p is the two-element sign subgroup and acts freely in the lifted fibre while disappearing after projection.

This closes a large class of earlier ambiguities. There is no longer a question about whether the ordinary frame carrier exists intrinsically, whether the visible action is torsor-like, whether the lifted carrier possesses its own internal action, or whether the two sheets are the restriction of the full internal group action. Yet (1) still lives over one point. A spin structure in established geometry is not a decoration of one fibre. It is a global relation between bundles over a base [5].

Task 26 therefore asks a more primitive question than “does a spin structure exist?”:

What minimum data are needed even to let the already reconstructed one-fibre frame geometry vary over a base, and which apparently familiar bundle properties actually follow from those data?

The answer is unexpectedly sharp. Fibrewise linear algebra and even algebraic transition functions can be reconstructed without difficulty. The first genuinely new obstruction is not algebraic but topological.

Methodological principle 1 (Change the problem class before changing the vocabulary). Once the one-fibre algebra is closed, passing to a base is not a notational generalization. It introduces a new dependency class: coherence of variation. A family of fibres must not be called a bundle until the topology or equivalent local-product structure has been supplied and verified.

2 Formal source, scope, and provenance

The supplied Task-26 archive has SHA-256

d5e793073d1129076b610c4f78b8ae8044f78366b4d9342d4d65833b03c144f8.

The pinned environment is Lean 4.28.0 with Mathlib revision

8f9d9cff6bd728b17a24e163c9402775d9e6a365.

The complete new dependency chain is

FamilyBase $\rightarrow$ FiberFrames $\rightarrow$ FrameTotal $\rightarrow$ FiberIsometryGroup $\rightarrow$ LocalTrivialization $\rightarrow$
FrameTrivialization $\rightarrow$ TransitionData $\rightarrow$ OneFiberComparison $\rightarrow$ Task26.

All earlier sources are reported unchanged.

Check	Result	Detail
Focused Task-26 endpoint build	PASS	endpoint target NullSectorTask26.Task26
Whole-project build	PASS	8298 jobs, 0 errors

Check	Result	Detail
Task-26 warnings	0	114 project-wide warnings are inherited from older unchanged layers
New source	9 modules	1615 Lean lines
Audited endpoints	46	all exposed through <code>Task26.lean</code>
Axiom baseline	exact	<code>propext</code> , <code>Classical.choice</code> , <code>Quot.sound</code>
<code>sorry/admit</code>	0/0	no proof holes
Prohibited escapes	0	no custom axiom, <code>implemented_by</code> , <code>native_decide</code> , <code>bv_decide</code> , <code>unsafe</code> , or <code>partial</code>
Inherited source	unchanged	Tasks I–XXV untouched in the supplied run

Proof boundary 1 (Verification boundary). *The source modules, endpoint file, audit, toolchain, manifest, archive hash, declaration statements, line counts, prohibited-token scan, and dependency structure were inspected for this paper. The reported Lean builds and `#print axioms` outputs are taken from the supplied formal artifact; the paper-generation environment did not independently rebuild the project.*

The scope firewall is essential. Task 26 introduces no manifold, tangent bundle, vector bundle as a primitive, principal bundle, lifted frame family, characteristic class, connection, curvature, or spin structure. Topology is introduced only in two negative-control statements showing that the automatically available topology is insufficient.

3 The minimal varying metric-oriented family

Let B be an arbitrary type and let $E : B \rightarrow \text{Type}$ be a dependent family. In the formal source, the varying-fibre input is

```
structure MetricOrientedRankThreeFamily
  (B : Type u) (E : B -> Type v)
  [forall b, NormedAddCommGroup (E b)]
  [forall b, InnerProductSpace R (E b)] where
  rank_three : forall b : B, finrank R (E b) = 3
  orient      : forall b : B, Orientation R (E b) (Fin 3)
```

Thus every fibre has a real inner-product-space structure, dimension three, and an orientation o_b . The base itself carries no topology at this stage.

The use of Mathlib’s `Orientation` is not cosmetic [8]. It prevents orientation from being encoded by a hidden chosen frame. Orientation preservation for a linear isometry $f : V \simeq W$ is defined by

$$\text{Orientation.map}(f)(o_V) = o_W. \quad (2)$$

Task 26 proves that this relation is stable under identity, inverse, and composition, and proves the determinant characterization

$$f \text{ preserves orientation} \iff \det(f) > 0 \quad (3)$$

in the relevant finite-dimensional setting.

Several negative controls prevent the data from collapsing into one another:

- rank three plus metric does not choose an orientation;
- the same metric family admits distinct orientations;
- an orientation does not choose a particular oriented orthonormal frame;
- data chosen at one base point do not determine data at another.

These are small results, but they enforce the dependency DAG: dimension, metric, orientation, and pointwise choice are different mathematical inputs.

4 Fibrewise frames and the total ordinary frame carrier

For an oriented inner-product space (V, o) , Task 26 defines

$$\mathcal{F}^+(V, o) := \{v : \text{OrthonormalBasis}(\text{Fin } 3, \mathbb{R}, V) \mid \text{orientation}(v) = o\}. \quad (4)$$

For the family this becomes

$$\mathcal{F}_b^+ := \mathcal{F}^+(E_b, o_b). \quad (5)$$

Every element is literally an orthonormal basis equipped with the proof that its basis orientation is the given orientation. No visible group and no reference frame occurs in the definition.

The formal development proves that each frame is a basis, is linearly independent, spans the whole fibre, and is extensional in its three vectors. More importantly, every $\mathcal{F}_b^+$ is nonempty. This uses the classical existence of an orthonormal basis followed, where necessary, by adjustment to the prescribed orientation.

The total ordinary-frame carrier is then merely the dependent sum

$$\mathcal{F}_{\text{tot}}^+ := \sum_{b:B} \mathcal{F}_b^+, \quad q : \mathcal{F}_{\text{tot}}^+ \rightarrow B, \quad q(b, F) = b. \quad (6)$$

Task 26 proves that the fibre of q over b is exactly $\mathcal{F}_b^+$ and that q is surjective.

Proof boundary 2 (Carrier versus bundle). *Equation (6) constructs a total set/type and a projection. It does not construct a frame bundle. Local product topology, continuity, and principal action data have not yet been supplied.*

This distinction is not terminological caution. It becomes a theorem-level issue in section 10.

5 The intrinsic fibre group and why a constant group is not canonical

For each oriented fibre Task 26 defines the subgroup

$$G_b^+ := \{g : E_b \simeq_{\text{lin.isom}} E_b \mid g \text{ preserves } o_b\}. \quad (7)$$

This group acts on $\mathcal{F}_b^+$ by mapping every vector in the frame. The action is proved free and transitive. Equivalently,

$$\forall F, G \in \mathcal{F}_b^+, \quad \exists! g \in G_b^+ : g \cdot F = G. \quad (8)$$

So every individual fibre has exactly the same torsor pattern reconstructed in the previous one-fibre stage.

The nontrivial point is how different fibres are compared. Choose a fixed model metric-oriented space (E_0, o_0) and define its group G_0^+ . An orientation-preserving linear isometry

$$\phi : E_b \simeq E_0 \quad (9)$$

induces by conjugation a group isomorphism

$$C_\phi : G_b^+ \xrightarrow{\sim} G_0^+, \quad C_\phi(g) = \phi g \phi^{-1}. \quad (10)$$

For another identification ψ , define

$$t_{\phi\psi} := \psi \phi^{-1} \in G_0^+. \quad (11)$$

The formal source proves the exact change law

$$C_\psi(g) = t_{\phi\psi} C_\phi(g) t_{\phi\psi}^{-1}. \quad (12)$$

It also proves that $C_\phi(g) = C_\psi(g)$ exactly when $t_{\phi\psi}$ commutes with $C_\phi(g)$.

Theorem 1 (No definitional constant structural group). *The metric-oriented rank-three family canonically supplies the intrinsic groups G_b^+ , but an identification of all of them with one fixed model group requires fibre-model identifications. Different identifications change the comparison by conjugation as in (12).*

This is already the standard mechanism by which a structural group appears through local coordinates rather than by declaring every fibre to be the same vector space.

6 Level 1: algebraic trivialization and the frame-as-chart equivalence

Task 26 deliberately splits local triviality into two levels. The closed Level 1 is purely algebraic.

For a subset $U \subseteq B$, a `LocalFibreTriv` is a family

$$\Phi = (\phi_b)_{b \in U}, \quad \phi_b : E_b \xrightarrow{\sim} E_0, \quad (13)$$

where every ϕ_b is a linear isometry and preserves orientation. No continuity in b is required.

A frame $F \in \mathcal{F}^+(V, o)$ itself determines an orientation-preserving isometric coordinate map

$$\text{chart}_F : V \xrightarrow{\sim} E_0 \quad (14)$$

by the representation map of the orthonormal basis. Conversely, an orientation-preserving metric identification with the model sends the model frame back to an oriented frame. The source proves an actual equivalence

$$\left\{ \phi : V \xrightarrow{\sim} E_0 \mid \phi \text{ preserves orientation} \right\} \simeq \mathcal{F}^+(V, o). \quad (15)$$

At family level this becomes

$$\text{LocalFibreTriv}(F, U) \simeq \prod_{b \in U} \mathcal{F}_b^+. \quad (16)$$

Thus the exact reverse-reconstruction theorem is not “every abstract product decomposition of the frame total space comes from a fibre trivialization”. It is the stronger and cleaner pointwise statement that a fibre-model identification is equivalent to a single oriented orthonormal frame, assembled over U .

Given Φ , Task 26 also constructs the induced full frame-space equivalence

$$q^{-1}(U) \simeq U \times \mathcal{F}^+(E_0, o_0) \quad (17)$$

at the set level. The direction from fibre charts to total frame charts is therefore complete. The converse for an arbitrary bare set equivalence of the form (17) is *not* claimed; an equivariance condition would be required to recover the underlying fibre maps.

7 The Choice collapse: Level 1 is globally trivial

The most important interpretation of (16) is easy to miss. Since every $\mathcal{F}_b^+$ is nonempty, classical choice produces one frame in every fibre. Consequently Task 26 proves

$$\forall U \subseteq B, \quad \text{Nonempty}(\text{LocalFibreTriv}(F, U)). \quad (18)$$

Taking $U = B$ yields a global *algebraic* trivialization.

Methodological principle 2 (Set-theoretic triviality is not geometric triviality). Once all fibres are known to be individually isometric to the same finite-dimensional model, choice can select those isometries independently point by point. Therefore any obstruction that survives in a genuine bundle problem must constrain how these choices vary, not whether they exist pointwise.

This changes the meaning of the transition data constructed next. At Level 1 they record the relative change between arbitrary pointwise coordinate choices, but they cannot yet encode a topological obstruction: one may simply choose a single global algebraic trivialization and eliminate all transitions. The difficulty has been pushed entirely into regularity of the variation.

8 Algebraic transition transformations

Let $\Phi = (\phi_b)$ and $\Psi = (\psi_b)$ be Level-1 trivializations over U and W . On an overlap point $b \in U \cap W$, Task 26 defines

$$g_{\Phi\Psi}(b) := \psi_b \phi_b^{-1} \in G_0^+. \quad (19)$$

Membership in G_0^+ is not assumed: it follows because both factors preserve the metric and orientation.

The three exact laws are proved:

$$g_{\Phi\Phi}(b) = 1, \quad (20)$$

$$g_{\Psi\Phi}(b) = g_{\Phi\Psi}(b)^{-1}, \quad (21)$$

$$g_{\Phi\Psi}(b) = g_{\Psi\Psi}(b) g_{\Phi\Psi}(b). \quad (22)$$

The order in (22) reflects the convention that group multiplication is composition: $(fg)(x) = f(g(x))$.

The frame trivializations transform by exactly the same group action:

$$\text{frameTriv}_\Psi(F) = g_{\Phi\Psi}(b) \cdot \text{frameTriv}_\Phi(F). \quad (23)$$

Hence the local coordinate ambiguity of fibres and the local coordinate ambiguity of frames are the same visible-group datum.

Methodological caveat 1 (What (22) does not yet mean). *The triple-overlap identity is already the algebraic cocycle law, but the maps $b \mapsto g_{ij}(b)$ have not been proved continuous because the required total-space topology has not been constructed. Thus the algebraic law is present before the topological bundle datum that would make it geometrically constraining.*

9 Comparison with the frozen one-fibre object

Task 26 explicitly connects the new family with the previously reconstructed one-fibre carrier without making that comparison canonical. A coordinate linear equivalence identifies the inherited positive-definite three-dimensional carrier with the fixed model and preserves the inner product. This yields an equivalence

$$\mathcal{F}^+(E_0, o_0) \simeq \mathcal{F}^+(E_{\text{old}}). \quad (24)$$

For a particular fibre E_b , any chosen orientation-preserving metric identification $\phi : E_b \simeq E_0$ then gives

$$\mathcal{F}_b^+ \simeq \mathcal{F}^+(E_{\text{old}}). \quad (25)$$

Changing ϕ changes (25) by the corresponding G_0^+ action. The previous one-fibre frame object is therefore a valid local model for every fibre, but no fibre is definitionally or canonically the old carrier.

This is the precise point at which the project has moved from “one object with many coordinate descriptions” to “many objects locally comparable with one model”.

10 The topological negative result

The natural temptation is to put the automatically available sigma topology on the total fibre type

$$E_{\text{tot}} := \sum_{b:B} E_b \quad (26)$$

and continue. Task 26 tests this rather than assuming it.

Suppose now that B has a topology. The formal source proves:

1. every individual fibre $\{(b, x) : x \in E_b\} \subseteq E_{\text{tot}}$ is open in the automatic sigma topology;
2. for *every* algebraic family of fibrewise continuous equivalences $f_b : E_b \simeq E_0$, the total map

$$(b, x) \longmapsto (b, f_b(x)) \quad (27)$$

is automatically continuous.

The second theorem uses Mathlib’s sigma-continuity characterization [9]. It is a negative control of unusual force: the topology makes every independent pointwise choice look continuous, so continuity expresses no relationship between nearby fibres.

For a non-discrete base this cannot be the intended locally product topology of a vector bundle. In a genuine local product $U \times E_0$, a single fibre over b is open exactly when the singleton $\{b\}$ is open in U ; the automatic sigma topology instead separates all fibres from the start.

Theorem 2 (Topological frontier). *The metric-oriented rank-three family, its frame total carrier, the algebraic local trivializations, and the transition laws do not determine the intended total-space topology used by bundle geometry. The automatically available sigma topology is formally too fine and makes the regularity requirement vacuous.*

This result is the principal new obstruction of Task 26. Unlike earlier no-go results, it does not forbid the desired structure. It identifies an additional datum that has not yet been reconstructed.

11 Comparison with established bundle mathematics

Classically, a rank- n real topological vector bundle over B is not merely an assignment $b \mapsto E_b$. It is a total space E_{tot} , a projection $E_{\text{tot}} \rightarrow B$, continuous fibrewise linear structure, and an open cover on which the total space is homeomorphic to $U_i \times \mathbb{R}^n$ by fibrewise linear maps [7, 3]. For a metric and oriented rank- n bundle, one can reduce local frames to positively oriented orthonormal frames, and the transition maps then take values in the orientation-preserving orthogonal group. The resulting oriented orthonormal frame space is naturally a principal group bundle once the topology and actions are in place [4].

Task 26 reconstructs the following parts of this standard picture without assuming the bundle itself:

- the base type and varying rank-three inner-product fibres;

- fibrewise orientation as independent data;
- fibrewise positively oriented orthonormal frame spaces;
- the total frame carrier and projection as a dependent sum;
- the intrinsic orientation-preserving isometry group of each fibre;
- free and transitive fibrewise group actions;
- model identifications and their conjugation dependence;
- algebraic local fibre trivializations;
- induced frame trivializations;
- transition transformations valued in the model visible group;
- identity, inverse, and triple-overlap laws.

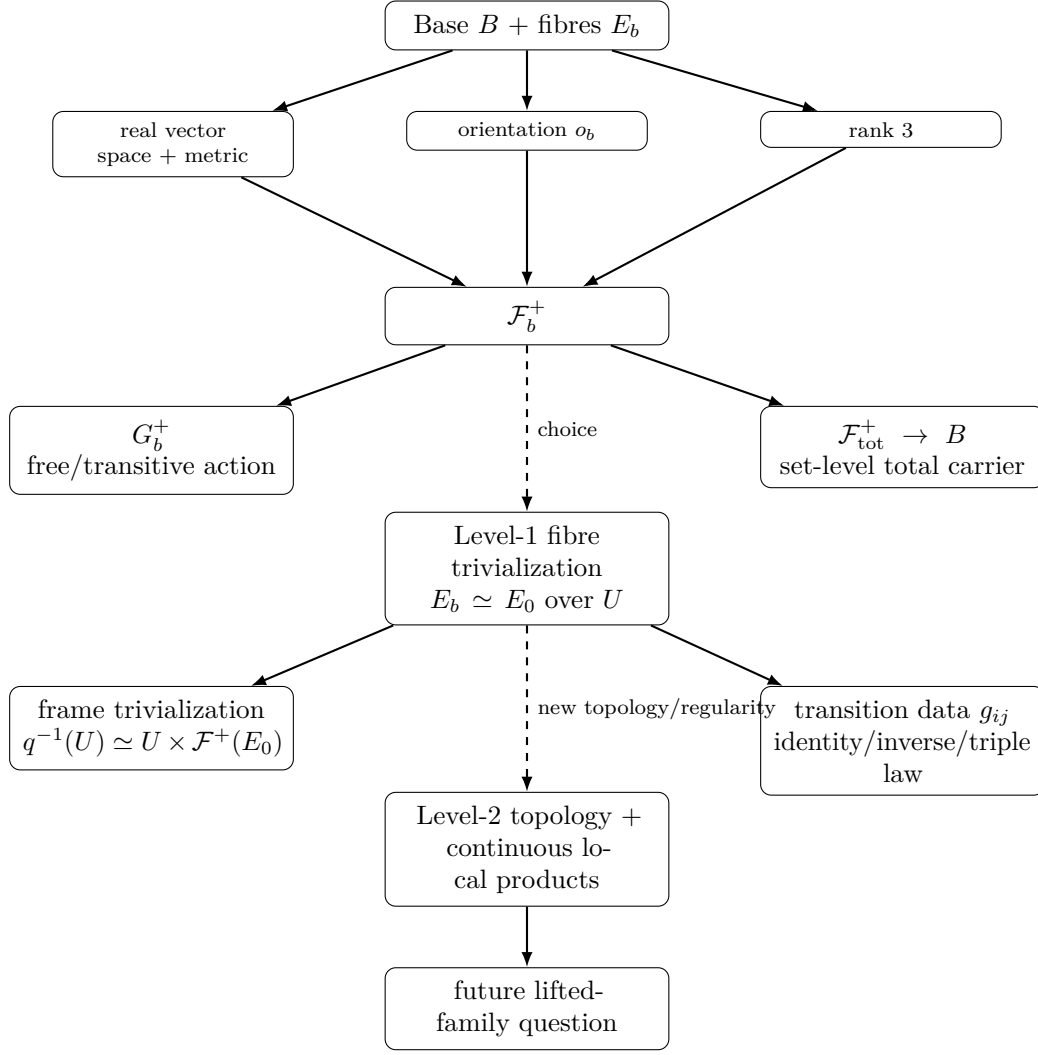
What remains absent is exactly what the classical definition adds beyond a set-family: topology, local product regularity, continuity of the structural operations and transition maps, and consequently the topological principal-bundle interpretation.

A spin structure is later still. In standard terms it is a principal lift of the oriented orthonormal frame bundle through the relevant two-sheeted group homomorphism, with an equivariant map covering the identity on the base [5]. The project already possesses the corresponding one-fibre group and torsor data, but Task 26 has not yet constructed the ordinary topological frame bundle to be lifted.

Status 1 (Reconstruction status after Task XXVI). The project has reached the algebraic precursor of an oriented metric frame bundle, not the bundle itself. The remaining gap to a spin structure is no longer pointwise: it is the global topology and the compatibility of local data across the base.

12 The exact dependency DAG

The logical structure exposed by Task 26 can be summarized as follows. Solid arrows are constructions/theorems; dashed arrows denote additional data rather than consequences.



The dashed arrow from Level 1 to Level 2 is the central discovery. Metric-oriented fibre data do not imply local topological triviality. Conversely, Task 26 shows that the algebraic portion below that arrow can be completed without any topology at all.

13 Engineering provenance and failed builds

The Task-26 audit records eleven intermediate compiler/proof failures. None changed a mathematical statement. They are retained because they distinguish implementation friction from mathematical obstruction.

#	Location	Diagnosis	Repair
1	FamilyBase	<code>congr</code> 1 already closed an orientation-transport goal; a trailing tactic saw no goals	removed the redundant trailing step
2	FamilyBase	reflexive linear-isometry equivalence was not syntactically the same as <code>LinearEquiv.refl</code>	inserted the explicit reflexive bridge
3	FamilyBase	orientation determinant theorem used a reindexed determinant	restated positivity with the required reindexing and used <code>Basis.det_reindex</code>

#	Location	Diagnosis	Repair
4	<code>FiberFrames</code>	orthonormal-basis projections required noncomputability	opened a noncomputable section
5	<code>FiberFrames</code>	generated <code>ext_iff</code> name collided with a manual lemma	renamed the manual lemma
6	<code>FiberFrames</code>	simplification normalized orthonormality/span expressions to a different syntactic form	used the explicit orthonormality characterization and basis-coercion rewrite
7	<code>FiberIsometryGroup</code>	uniqueness proof used the wrong intermediate group identity	reorganized the proof through $g^{-1} \cdot w = v$
8	<code>FiberIsometryGroup</code>	no scalar-action instance was found through non-reducible fibrewise definitions	added the fibrewise <code>MulAction</code> instances
9	<code>FrameTrivialization</code>	unresolved model basis-vector goals after simplification	added the explicit frame-chart inverse-on-single lemma
10	<code>TransitionData</code>	model group action instance was absent under the model-group name	added <code>modelGvisMulAction</code>
11	<code>Task26</code>	compiler IR failed on aliases of noncomputable definitions	replaced three aliases by noncomputable definitions

All eleven repairs are classified by the supplied audit as proof/elaboration engineering only. The final theorem boundary is unchanged.

14 Why a spin structure is still absent

The amount reconstructed by this point can make the remaining gap look deceptively small. It is not. The project now has, fibrewise, the ordinary frame torsor and the previously certified lifted one-fibre torsor. It also has an algebraic atlas-like family of coordinate choices and visible transition transformations. But the following objects remain absent:

1. a correct topology on the total vector-space carrier $\sum_b E_b$;
2. a topology on the total frame carrier compatible with the fibre topology and the projection;
3. local trivializations that are homeomorphisms rather than pointwise isometries selected independently;
4. continuity of vector-space operations across the total space;
5. continuity of frame trivializations and transition maps;
6. a demonstrated topological principal action on the ordinary frame total space;
7. a lifted total space over the same base;
8. compatible lifted transition data and their global gluing.

Until the first six items exist, the last two cannot be asked in the correct category.

Proof boundary 3 (Pointwise versus global). *The one-fibre double structure solved earlier is a necessary local model for the later global geometry, but it does not provide the missing topology of a varying family. Pointwise equivalence of all fibres with a model is strictly weaker than coherent local equivalence over open subsets.*

15 Part XXVII: the topology of variation

The formal outcome of Task 26 changes the most economical next step. The transition maps already satisfy all algebraic overlap laws, but attempting to lift them immediately would ask a regularity question in a setting where regularity is not yet defined correctly. Part XXVII should therefore harden the ordinary family first.

A tightly focused next task should investigate at most three hard targets.

15.1 A correct total-space topology

The first question is whether the topology should be taken as additional structure on E_{tot} or reconstructed from an atlas of local product identifications. A successful formulation must force the projection $E_{\text{tot}} \rightarrow B$ and the fibrewise linear operations to be continuous while restricting to the existing topology on each fibre. It must explicitly exclude the already refuted sigma/disjoint-union topology.

15.2 Level-2 local triviality

For open $U \subseteq B$, the algebraic maps

$$q^{-1}(U) \longrightarrow U \times E_0 \tag{28}$$

should become homeomorphisms whose restrictions to fibres are the orientation-preserving linear isometries already formalized. The difficult dependency question is two-way:

Does a compatible local atlas determine the total-space topology uniquely, and conversely does a suitable total-space topology plus local triviality recover the atlas data used in Task 26?

This is not a theorem of Task 26 and should not be assumed in advance.

15.3 Continuous transition data and the ordinary frame bundle frontier

Once Level 2 is available, the existing pointwise transformations $g_{ij}(b)$ should be tested for continuity. The ordinary frame total carrier should then inherit local product charts with fibre $\mathcal{F}^+(E_0, o_0)$ and, only if the required continuity and action properties are proved, may be identified as a topological principal frame object.

At that point a later task can ask a new question about the already certified two-sheeted internal projection: whether the continuous visible transition system admits locally compatible internal representatives and what exact overlap condition controls their gluing. Task XXVII itself should stop before solving that problem. In this way the next stage remains reconstruction-first: the ordinary topological carrier is completed before any additional lifted structure is imposed.

Open question 1 (The new hard frontier). *Which topology and local-product axioms are minimally sufficient to convert the algebraic family of Task 26 into a genuine topological metric-oriented rank-three bundle, without importing the completed bundle object as a primitive?*

16 Conservation of Difficulty

The sequence of difficulty movements is now unusually transparent.

1. Earlier stages exposed the internal/visible two-sheeted group structure.
2. The local cover was reconstructed from local representatives and sign-valued transition data.
3. The ordinary frame carrier was then reconstructed intrinsically and shown to be a visible-group torsor.
4. The lifted frame carrier was hardened into an internal-group torsor with equivariant projection.
5. Task 26 let these objects vary pointwise over a base and recovered the full algebraic transition law.
6. At that point algebra became too weak to contain the remaining difficulty: choice makes every such pointwise family algebraically trivial.
7. The difficulty therefore reappears as topological coherence of variation.

This is a particularly clean instance of the Conservation of Difficulty principle. Nothing mysterious remains hidden in a single fibre. The unresolved content is exactly the information that is lost when neighbouring fibres are treated as independent sets rather than as parts of one topological total space.

17 Decision table and theorem boundary

Question	Formal verdict
Arbitrary base without manifold assumptions?	PROVED / built into the family definition
Varying real rank-three fibres?	PROVED
Fibrewise positive-definite metric?	PROVED / inherited from <code>InnerProductSpace</code>
Fibrewise orientation independent of a frame?	PROVED
Intrinsic $\mathcal{F}_b^+$?	PROVED
Frames exist in every fibre?	PROVED, classically
Total ordinary-frame carrier?	PROVED
Its fibre over b is exactly $\mathcal{F}_b^+$?	PROVED
Intrinsic orientation-preserving fibre group?	PROVED
Free/transitive fibrewise action?	PROVED
Canonical identification of every fibre group with one fixed group?	REFUTED as canonical; requires a model identification and changes by conjugation
Algebraic local fibre trivialization?	PROVED
Existence over every subset, including the whole base?	PROVED by classical choice
Induced full frame-space trivialization?	PROVED at set level
Reverse reconstruction from one frame section?	PROVED, existence and uniqueness
Arbitrary frame-total-space product equivalence classified?	NOT PROVED; would require extra equivariance

Question	Formal verdict
Transition transformations in the model group?	PROVED
Identity/inverse/triple-overlap laws?	PROVED
Continuity of transition transformations?	NOT YET STATABLE in the intended topology
Automatic sigma topology suitable?	REFUTED by two negative controls
Correct bundle topology constructed?	NO
Topological frame bundle constructed?	NO
Principal bundle constructed?	NO
Lifted frame family constructed?	NO
Spin structure constructed?	NO

18 Conclusion

Part XXVI reaches the first genuinely global frontier of the reconstruction. Starting from an arbitrary base and varying metric-oriented rank-three fibres, it derives the intrinsic frame carriers, their fibrewise orientation-preserving isometry groups, the free/transitive actions, the total ordinary-frame carrier, algebraic local trivializations, exact frame reconstructions, and visible transition transformations with their full overlap laws. It also proves that comparison with one fixed visible model is choice-dependent by conjugation.

The apparently negative result about topology is the decisive advance. Since algebraic trivializations exist globally by choice, the remaining geometry cannot be located in pointwise existence. Since the automatic sigma topology makes every fibre open and every pointwise algebraic trivialization continuous, the intended coherence cannot be obtained for free from the dependent type either. A new mathematical dependency has been isolated: the topology of variation.

The project is therefore closer to a spin structure in one sense and farther from a shortcut in another. The pointwise group and frame architecture is increasingly complete, but a spin structure remains impossible to claim until the ordinary varying frame object has become a genuine topological bundle. Part XXVII should reconstruct precisely that missing Level-2 layer and stop there. Only after the ordinary transition system is genuinely continuous will the next lifted gluing question be mathematically well-posed.

A Task-26 principal theorem map

The 46 audited endpoints fall into the following mathematical groups:

1. family and orientation infrastructure: `MetricOrientedRankThreeFamily`, finite dimensionality, orientation-preservation composition/inverse/identity, determinant characterization;
2. dependency negative controls: two orientations, same metric with different orientations, multiple oriented orthonormal bases, pointwise data not global;
3. fibre frames: the intrinsic frame carriers together with basis, independence, span, extensionality, and nonemptiness results;
4. total frames: `FrameTotal`, `frameBase`, exact fibre equivalence, surjectivity, sigma-fibre openness;
5. fibre groups: `S00f`, `GvisAt`, free/transitive action, unique transporter;
6. model comparison: `gvisCompare`, change by conjugation, equality iff commutation;

7. Level-1 trivialization: `LocalFibreTriv`, construction from frames, existence over arbitrary subsets;
8. frame reconstruction: `frameChart`, orientation preservation, `frameEquivOfIsom`, total frame trivialization, uniqueness/existence of isometry from frame image, equivalence between model identifications and frames, family-level trivialization/frame-section equivalence;
9. transition data: transition definition, metric/orientation preservation, identity, inverse, triple law, frame-transition compatibility;
10. comparison with the frozen one-fibre carrier: coordinate equivalence, model frame equivalence, fibre-to-old-frame equivalence, exact dependence on model identification;
11. frontier interface: `LiftedFamilyFrontier` and its inherited transition laws, with no lifted object in the record.

B Exact unresolved obligations after Task XXVI

The supplied audit lists the following unresolved mathematical obligations, retained here without promotion to established results:

1. construct or specify the intended topology on the total fibre carrier and on the frame total carrier;
2. define Level-2 topological local triviality and compare it precisely with Level 1;
3. prove continuity/regularity of the transition transformations once the correct topology exists;
4. characterize which equivariant frame-total-space trivializations reconstruct fibre trivializations; arbitrary bare set equivalences are not covered;
5. extend the minimal use of the orientation API only if future topological constructions require it;
6. only after the ordinary topological family is closed, formulate a lifted-family/gluing problem; no principal bundle or spin structure is yet present.

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